3/25/2022

Friday, March 25, 2022 10:45 AM

M168_Notes_7.5

7.5 Conditional Probability and Independent Events

A survey is done of people making purchases at a gas station. Most people buy gas (Event A) or a drink (Event B).

	buy drink (B)	no drink (B^c)	total
buy gas (A)	45	25	70
no gas (A^c)	20	10	30
total	65	35	100

What is the probability that a person bought gas and a drink?

$\frac{45}{100}$

What the probability that *a person who buys a drink* also buys gas? In other words, **given** that a person *bought a drink* (B), what is the probability that they bought gas (A)?

$\frac{45}{65}$

NOTATION: P(A|B) = the probability of A given B

The ***conditional probability*** of event E given event F is

$$P(E \mid F) = \frac{n(E \cap F)}{n(F)} = \frac{n(E \cap F)/n(S)}{n(F)/n(S)} = \frac{P(E \cap F)}{P(F)}$$

What is the probability that *a person who buys gas* also buys a drink?

PRODUCT RULE: $P(E \cap F) = P(E)P(F|E)$

TREE DIAGRAMS

At a party, 1/3 of the guests are women. 75% of the women wore sandals and 25% of the men wore sandals. What is the probability that a person chosen at random at the party is a man wearing sandals, $P(M \cap S)$? What is the probability that a randomly chosen guest is wearing sandals?

A finite stochastic process is one in which the next stage of the process depends on the state you are in for the previous stage.

Consider drawing 3 cards from a standard deck of 52 cards without replacement.
(a) What is the probability that the three cards are hearts?
(b) What is the probability that the third card drawn is a heart given the first two cards are hearts?

INDEPENDENT EVENTS: P(E|F) = P(E)

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)} = P(E) \quad \Rightarrow$$

$$P(E \cap F) = P(E) \cdot P(F) \textit{ iff } E \text{ and } F \text{ are independent}$$

<u>*Example*</u>

A certain part on airplane has a 1% chance of failure. If they carry two back-ups to this part, what is the probability that they will all fail?

<u>*Example*</u>

A medical experiment showed the probability that a new medicine was effective was 0.75, the probability of a certain side effect was 0.4 and the probability for both occurring is 0.3. Are these events independent?